

Process Characterization Plan

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCP-004

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCP-004
Document class	Process Characterization Plan
Title	Process Characterization Plan: Harvest and Clarification (Step 4)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; BLA, Biologics License Application; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; NTU,

nephelometric turbidity unit; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal force; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure.

1 Purpose and scope

Harvest and clarification is the primary recovery step of the A-Mab drug substance process, and this plan defines the studies that will characterize it. The step separates the antibody from cells and cell debris, and it delivers the clarified harvest that the Protein A capture step (Step 5) receives. It forms no product quality attribute of its own, and its parameters exist for two purposes: to make the product recovery reproducible, and to keep the composition of the material handed to capture consistent from batch to batch.

The scope covers the three parameters listed in Table 6.1, studied at a qualified scale-down model of the commercial harvest train (SOP-1001). Centrifugation intensity (rcf) and depth filter load are the two settable inputs. Post-clarification turbidity is the attribute by which the clarified harvest is released to capture, and its range is in scope because it defines the feed that the capture step has to accept.

The plan does not cover the fed-batch culture that produces the broth, which is characterized under PCP-003. It does not cover the capture step, which is characterized under PCP-005. No viral clearance is claimed for harvest and clarification, and no log reduction will be measured across it.

The work is Stage 1 process design under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and it follows the enhanced development approach of ICH Q8, ICH Q9 and ICH Q11 (International Council for Harmonisation 2009, 2023, 2012). The methodology, the attribute framework and the parameter classification scheme are those of the A-Mab case study (CMC Biotech Working Group 2009). The studies support the transfer of the drug substance process from the sending site to the receiving site under PTP-001.

2 Objectives

The study has five objectives.

- To establish the effect of centrifugation intensity on centrate clarity, on product recovery, and on the impurity burden released into the centrate by cell lysis (HCP and residual DNA).
- To establish the effect of depth filter load on filtrate turbidity, on the differential pressure of the filter train, and on product recovery.
- To confirm that the size variant profile of the antibody (high molecular weight species by SEC-HPLC) does not change across the step over the ranges studied.
- To characterize the host cell protein (HCP) and residual DNA burden of the clarified harvest, which is the input the capture step has to clear.
- To propose a proven acceptable range and a normal operating range for each parameter, and to support the classification of each parameter under SOP-4001.

3 Related documents

Table 3.1 lists the documents that govern this plan and the documents that depend on it. The controlled procedures and the validated analytical methods used in the study are listed in Table 3.2.

Table 3.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-004	Process Characterization Report (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2: Controlled procedures and validated analytical methods for the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Harvest and clarification is a platform operation at Novacyte Biologics, so the study confirms and bounds behaviour that is already understood. The same continuous disk-stack centrifuge and the same graded depth filter train have been used for three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and for the A-Mab clinical campaigns. Across those products the step has consistently delivered clarified harvest of acceptable turbidity and consistent recovery. This experience reduces the work required here to a confirmation of the proposed ranges at the commercial scale-down model.

Table 4.1 places the step in the drug substance train. Step 4 sits between the production bioreactor (Step 3), which forms the glycan, charge variant and aggregate attributes, and the capture step (Step 5), which begins the removal of process impurities. The clarified harvest is the only material transferred between them.

Table 4.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The two settable parameters act through different mechanisms, and each one carries a penalty at the end of its range. Centrifugation removes cells and large debris by

sedimentation, so a higher relative centrifugal force gives a clearer centrate. Higher force also raises the hydrodynamic shear in the feed zone and at the solids discharge (the two places at which the broth is accelerated most sharply), which breaks a fraction of the cells and releases HCP, DNA and fine particles into the centrate. The two effects oppose each other, and the set-point is expected to lie near the force at which the gain in clarity no longer pays for the additional lysis.

Depth filtration retains the fine particles that the centrifuge does not remove. Its capacity is expressed as the volume filtered per unit of media area (L/m²), and both the filtrate turbidity and the differential pressure rise as that load increases. Loading beyond the capacity of the primary media allows fines to break through into the filtrate, which raises the turbidity of the clarified harvest and increases the particulate burden carried into the capture column, and both of those effects are expected at the high end of the load range studied here. The depth filter media also adsorb a fraction of the HCP and the DNA in the centrate (an effect seen across the platform products). That clearance varies with the state of the culture at harvest and is not predictable, so none is claimed for this step.

4.2 Quality attributes in scope

The step sets no critical quality attribute. The attributes in scope are the ones it carries forward to the purification train (Table 4.2), and they are given there with their drug substance acceptance criteria and their criticality.

Table 4.2: Quality attributes carried forward from harvest and clarification. The step forms none of them and is credited with no clearance of them.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60

Host cell protein is the attribute in this table that most constrains the design of the downstream train. The clarified harvest is expected to carry about 500,000 ng/mg of HCP against a drug substance limit of 100 ng/mg, so the purification train has to deliver about 3.7 logs of clearance in total, of which the capture step provides the largest single contribution (PCR-005) and the two ion exchange polishing steps provide the remainder (PCR-007 and PCR-008). Harvest and clarification is credited with none of it. What the step does control is the consistency of the burden it hands over, which is why HCP and residual DNA are measured on the clarified harvest at every studied condition.

The size variant profile is measured for a different reason. Shear at the centrifuge feed zone is the only mechanism at this step that could raise the level of high molecular weight species, and the study measures aggregate across the step to confirm that it does not. Aggregate reduction is the business of the cation exchange step (PCR-007), and none is expected or claimed here.

4.3 Risk-based prioritization of parameters

RA-001 assigned a study type to every parameter in the drug substance train (37 parameters in total), and all three harvest parameters were assigned to univariate study. Neither settable parameter is linked to a critical quality attribute, and no interaction between them is expected to act on product quality, so neither reached the risk score at which a multivariate design is required. The assigned study type is carried into the last column of Table [6.1](#).

The characterization ranges are wider than the routine control ranges, so that the study bounds the consequence of an excursion as well as the behaviour at the set-point. The centrifugation range spans 6000 to 12000 g against a normal operating range of 8000 to 10000 g. The depth filter load range spans 80 to 160 L/m² against a normal operating range of 100 to 140 L/m². Turbidity is studied to 20 NTU (AMV-3015), which is twice the upper edge of its normal operating range, because the capture step has to be shown to tolerate a harvest at the worst clarity that the clarification train can produce.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run at a scale-down model of the commercial harvest train, qualified before the first characterization run (SOP-1001). The model uses a bench continuous disk-stack centrifuge matched to the commercial machine on equivalent settling area and on feed residence time, followed by a depth filter train of the same media grades in the same sequence, operated at the same volume per unit of media area. Scale-independent settings (relative centrifugal force, load per unit area, temperature) will be held at the values proposed for commercial operation. Scale-dependent settings (feed flow rate, solids discharge interval, hold-up volumes) will be established so that the model matches the performance of the commercial train.

At commercial scale the filtration duty follows from the load parameter and the batch volume. A working volume of 15,000 L (the commercial bioreactor volume) at the set-point load of 120 L/m² calls for about 125 m² of depth filter media, and the normal operating range corresponds to about 107 to 150 m². The scale-down model is operated at the same volume per unit area, so the duty on the media is preserved even though the absolute area is not.

Qualification will use triplicate model runs on a common feed. The model results will be compared with the corresponding at-scale records for centrate turbidity, clarified harvest turbidity, step recovery, and the HCP and DNA content of the clarified harvest. The acceptance criteria for that comparison are given below (§7). The qualification report will be issued before the characterization runs begin and will be summarized in PCR-004.

The model has a known limitation. It reproduces the sedimentation and filtration duty of the commercial train, but it does not reproduce the residence time distribution of the commercial bowl or the frequency of its solids discharge. Confirmation at commercial scale comes from the turbidity and recovery records of the PPQ batches, and this limitation will be restated in the report.

5.2 Operation

The broth will be harvested at the end of the fed-batch culture, at the nominal end of culture viability of 67% for the platform process and at a nominal titre of 4.37 g/L (PCP-003). Broth is fed continuously to the centrifuge (SOP-2005). The centrate passes to the graded depth filter train and then to a sterilising grade filter (SOP-2006). The clarified harvest is collected, sampled and held for capture.

Two feed states will be used, because the solids load and the lysis-derived impurity burden both depend on the state of the culture at harvest. The first is a culture harvested at the nominal duration. The second is a culture harvested after an extended duration, which has lower viability, a higher solids load and a higher impurity burden, and which is the worst case that the clarification train has to handle. The culture conditions that produce the two feed states are defined in PCP-003.

5.3 Analytical methods

The validated methods are listed in Table 3.2. Turbidity in the centrate and in the clarified harvest is measured by nephelometry (AMV-3015). HCP is measured by ELISA (AMV-3012), residual DNA by qPCR (AMV-3014), and the size variant profile by SEC-HPLC (AMV-3011).

Method precision sets the smallest difference the study can resolve, so it is stated here and used again in the decision criteria of §7. The turbidity method has a precision of 4.0% and a limit of quantitation of 0.5 NTU. The upper edge of the turbidity operating range therefore sits about 20 times above the limit of quantitation, and the whole of the studied range is resolved. The HCP ELISA has a precision of 9.5%, and a difference in the clarified harvest burden smaller than that will not be separated from assay variability. Full method performance is reported in the validations listed in Table 3.2.

5.4 Sampling plan

Samples will be taken at four points in every run: the bioreactor broth before centrifugation, the centrate, the pooled depth filtrate, and the clarified harvest after the sterilising filter (the material released to capture). Turbidity is measured at every point. HCP, residual DNA and the size variant profile are measured on the broth and on the clarified harvest, so that the change across the whole step is measured directly and not inferred from the values recorded at the two intermediate points. Volume and protein concentration are recorded at each point for the mass balance, and the differential pressure across the depth filter train is recorded continuously.

5.5 Statistical methods

The analysis is univariate, because no parameter at this step is linked to a critical quality attribute. Each parameter will be varied across its characterization range while the other parameters are held at their set-points, and each response will be summarized as a mean and a standard deviation over the runs executed at that condition on each of the two feed states. The replicated set-point condition provides the reproducibility term against which the edge conditions are compared.

Edge conditions will be compared with the set-point condition by a two-sided t-test ($\alpha = 0.05$). Statistical significance will be reported separately from practical significance. A difference that is significant but smaller than the precision of the method that measured it will be reported as detected and not meaningful, with the method precision given alongside it.

No response surface model will be fitted, and no design space will be claimed for this step. Process capability for the attributes in Table 4.2 is established at the steps that govern them (PCR-005, PCR-007 and PCR-008) and consolidated in PCMR-001.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters, their set-points, the ranges to be studied, the normal operating ranges and the study type. The centrifugation set-point of 9000 g and the depth filter load set-point of 120 L/m² are the platform values proposed for commercial operation, and each one is the centre of its characterization range.

Table 6.1: Parameters to be studied, with set-points, characterization ranges, normal operating ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Centrifugation (rcf)	g	9000	6000–12000	8000–10000	univariate
Depth filter load	L/m ²	120	80–160	100–140	univariate
Post-clarification turbidity	NTU	5	1–20	1–10	univariate

Post-clarification turbidity is not set directly. It is the outcome of the centrifuge and the depth filter train, and it is the attribute by which the clarified harvest is judged before it is fed to capture. It appears in Table 6.1 because its range defines the feed that the capture step has to accept, and because it is the in-process control that will be applied in routine manufacture.

6.2 Univariate assessment

Each settable parameter will be studied at its low characterization edge, at its set-point and at its high characterization edge, with the other settable parameter held at its set-point, which gives 6 clarification conditions in all, listed in Table 6.2. Every condition will be run on both feed states. The set-point condition will be run in triplicate on each feed state, and it is the reference for every comparison.

Table 6.2: Planned univariate conditions. Each condition is run on both feed states; the set-point condition is run in triplicate.

Parameter varied	Unit	Low edge	Set-point	High edge	Other parameter held at
Centrifugation (rcf)	g	6000	9000	12000	Depth filter load at 120 L/m2
Depth filter load	L/m2	80	120	160	Centrifugation (rcf) at 9000 g

Two corners of the studied region are worst cases, for opposite reasons. The low centrifugation and high filter load corner carries the largest solids burden into a filter train that is already near the end of its capacity, and it is expected to give the highest turbidity in the clarified harvest and the highest differential pressure across the depth filters. The high centrifugation corner gives the clearest centrate but the greatest shear, and it is the condition at which the lysis-derived HCP and DNA burden is expected to be highest. Both corners will be run on the extended culture feed state.

The turbidity range will be spanned by these conditions, and, where the conditions do not reach the edges of the range, by blending clarified harvest to the required turbidity. Material at the low, set-point and high end of the turbidity range will be retained for the capture feed challenge described in PCP-005.

7 Acceptance and decision criteria

Table 7.1 gives the criteria that each studied condition has to meet. Three of them are process performance criteria (turbidity, recovery and filter train pressure). The other two confirm that the step does not alter product quality and that the impurity burden handed to capture is characterized.

Table 7.1: Acceptance and decision criteria for a studied condition.

Attribute or outcome	Method or basis	Criterion for a studied condition
Post-clarification turbidity	Nephelometry (AMV-3015)	Within the normal operating range of 1 to 10 NTU at the set-point condition; reported without a limit at the other conditions
Product recovery over the step	Mass balance from volume and protein concentration	Mean not below 89.3% of the product mass entering the step
Depth filter train pressure	In-process differential pressure	Batch volume completed without exceeding the media manufacturer's rated differential pressure
Aggregates (HMW)	SEC-HPLC (AMV-3011)	No increase across the step beyond the precision of the method
Host cell protein and residual DNA	ELISA (AMV-3012); qPCR (AMV-3014)	Characterized and reported as the capture feed; no limit applied at this step

Product recovery is the criterion with the largest consequence for the process. The platform recovery over this step is 95.0% with a batch-to-batch coefficient of variation of 2.0%, so a condition will be judged to have reduced recovery when its mean falls below 89.3%, which is three standard deviations below the platform mean. Turbidity in the clarified harvest has to remain inside its normal operating range at the set-point condition, and it is reported without a limit at the other conditions.

A condition that fails a criterion does not invalidate the study. The proven acceptable range of that parameter is narrowed to the last level that met every criterion, and the narrowed range is carried into PCR-004 together with the evidence for it. A failure at the set-point condition, or a failure of the scale-down model qualification, stops the study and is escalated under the quality system.

The scale-down model qualification will be accepted when the model and at-scale values of centrate turbidity, clarified harvest turbidity and step recovery agree within the precision of the methods that measure them, and when the HCP and DNA content

of the clarified harvest produced by the model falls inside the range recorded at commercial scale. If the model fails on any of these, it will be modified and requalified before any characterization run is executed.

The study is complete when every condition in Table 6.2 has been executed on both feed states, every planned sample has been assayed, the mass balance has been closed, and the criteria in Table 7.1 have been evaluated for each condition.

8 Proven acceptable ranges (planned analysis)

The proven acceptable ranges for this step will be derived from process performance and not from a drug substance specification. No critical quality attribute is formed or cleared here, so there is no attribute limit from which a step level criterion could be back-calculated (as it is for the viral clearance steps). The acceptance basis is the set of criteria in Table 7.1, together with the requirement that the clarified harvest is acceptable as feed to the capture step.

For each settable parameter, the proven acceptable range will be the contiguous interval around the set-point over which every criterion in Table 7.1 is met on both feed states, and it will be read from the univariate series and reported next to the full characterization range so that the margin between the two is visible. Where every studied level meets the criteria, the whole characterization range will be reported as proven acceptable, and the binding constraint on routine operation will remain the normal operating range.

Two analyses that a multivariate step would report will not be run here. There is no fitted response surface model, so a proven acceptable range cannot be propagated by Monte-Carlo simulation of the other parameters varying within their normal operating ranges. There is no multivariate operating region either, so no design space is claimed for the step. The proven acceptable ranges reported for harvest and clarification will be univariate, and they will hold for the two feed states studied and at the qualified scale-down model.

Each proven acceptable range will be presented as a plot of the response against the parameter, with the acceptance criterion drawn, the acceptable region shaded green, and the set-point and the normal operating range marked. The plots and the data behind them will be given in the report (PCR-004).

9 Data management and integrity

Study data will be recorded and reviewed under the ALCOA+ principles. Raw analytical data are captured in the validated laboratory systems named in the method validations (Table 3.2), and process data are captured in the scale-down model batch records. Each record is reviewed by a second qualified person before the data are used in the report. Changes to an executed record are made by controlled amendment with the reason recorded.

10 Roles and responsibilities

Process Development prepares this plan, executes the scale-down runs and authors PCR-004. Analytical Development performs the assays and releases the results against the validated methods. MSAT owns the scale-down model qualification and the comparison with the at-scale records (§5.1). Statistics reviews the analysis and the decision rules. Quality Assurance approves the plan and the report and disposes any deviation.

11 Deliverables and schedule

The deliverable is PCR-004, the process characterization report for this step (Table [3.1](#)). The work runs in three phases. Scale-down model qualification is executed and reported first. The characterization runs on the two feed states follow, with the analytical testing and the mass balance closed out as each condition completes. The report is then authored, reviewed and approved, and it rolls up into PCMR-001. No dates are given in this synthetic plan.

12 Risks and assumptions

The main risk to the study is the representativeness of the scale-down model. The model is qualified against at-scale records before use, and its limitation (§5.1) is carried into the report.

The second risk is the supply of the extended culture feed state, which depends on the bioreactor campaign run under PCP-003, and if that material is not available the worst-case conditions will be executed on the nominal feed state and the gap stated in PCR-004 as a bound on the conclusions.

The study assumes that the platform centrifuge and the platform depth filter media are unchanged for commercial manufacture. It also assumes that the impurity burden of the clarified harvest is set by the culture and not by the clarification train. The HCP and DNA measurements taken across the step will test that second assumption directly.

13 Approvals

This plan takes effect when the approvals recorded in Table 1 are complete. Changes after approval are made by controlled revision. Any departure from the plan during execution is recorded at the time and assessed in PCR-004.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

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